

“数字集成电路设计” 2023-2024 第 1 学期 第 3 周作业

10%

- 6.1 有较低 V_{OL} 的反相器总有较短的从高到低的开关时间吗？给出理由证明你的回答。

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- 6.2 考察 $10\text{ k}\Omega$ 电阻负载的反相器电路对于 10 fF 电容的开关延迟，这里：

$$\mu_n C_{ox} = 98.3\text{ }\mu\text{A/V}^2$$

$$(W/L)_n = 10$$

$$V_{T,n} = 0.53\text{ V}$$

$$E_{c,n} L_n = 0.4\text{ V}$$

- a. 用平均电流法求 τ_{PHL} (从高到低下降到 50% 时的传输延迟)。假设输入信号是在 0 V 和 1.2 V 之间跳变的理想矩形脉冲，其上升和下降时间为 0 。必须计算 V_{OL} 来解这个问题。
- b. 用合适的微分方程和恰当的不为 0 的电容起始电压 V_{OL} (输入电压为 V_{OH}) 以及 (a) 中相同的输入电压来计算 τ_{PLH} 。

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- 6.8 某个 CMOS 反相器的器件参数如下：

$$\text{nMOS } V_{T0,n} = 0.5\text{ V} \quad \mu_n C_{ox} = 98\text{ }\mu\text{A/V}^2 \quad E_{c,n} L_n = 0.4\text{ V}$$

$$\text{pMOS } V_{T0,p} = -0.48\text{ V} \quad \mu_p C_{ox} = 46\text{ }\mu\text{A/V}^2 \quad E_{c,p} L_p = 1.8\text{ V}$$

电源电压 $V_{DD} = 1.2\text{ V}$ 。两个晶体管的沟道长度 $L_n = L_p = 40\text{ nm}$ 。电路总的输出负载电容 $C_{out} = 200\text{ fF}$ ，它与晶体管尺寸无关。

- a. 求 nMOS 和 pMOS 管的沟道宽度，使得开关门限电压为 0.59 V ，输出上升时间 $\tau_{rise} = 100\text{ ps}$ 。
- b. 计算 (a) 中电路的平均传播延迟时间 τ_P 。
- c. 如果电源电压从 1.2 V 降到 1.0 V ，开关门限电压 V_{th} 和延迟时间将如何变化？解释原因。

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- 6.9 考虑具有与习题 6.8 中相同工艺参数的 CMOS 反相器。开关门限电压设计为 0.58 V 。总输出负载电容的简化表达式如下：

$$C_{out} = 5\text{ fF} + C_{db,n} + C_{db,p}$$

而且，已知 nMOS 和 pMOS 管的漏极到衬底的寄生电容是沟道宽度的函数。一系列的简化电容表达式如下：

$$C_{db,n} = 0.16\text{ fF} + 1.7W_n$$

$$C_{db,p} = 0.13\text{ fF} + 1.4W_p$$

这里， W_n 和 W_p 的单位是 μm 。

- a. 求两个晶体管的沟道宽度，使传播延迟时间 τ_{PHL} 小于 35 ps 。
- b. 假设现在 CMOS 反相器的 $(W/L)_n = 10$ ， $(W/L)_p = 15$ ，总的输出负载电容是 5 fF 。用平均电流法计算输出上升和下降时间。

参考答案:

6.1 (10)

不一定。(4分)

以电阻负载反相器为例, 电阻的阻值增大, V_{OL} 变小, 然而延迟时间变大。(6分)

6.2(30)

a) (24分)

(短沟道方法)

当 $V_{out} = V_{OL}$ 时, 假设晶体管工作在线性区

$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{k_n}{2} \frac{1}{\left(1 + \frac{V_{OL}}{E_C L_n}\right)} \left[2 \cdot (V_{in} - V_{T,n}) \cdot V_{OL} - V_{OL}^2 \right]$$

$$\frac{1.2 - V_{OL}}{10 \times 10^3} = \frac{983 \times 10^{-6}}{2} \frac{1}{\left(1 + \frac{V_{OL}}{0.4}\right)} \left[2(1.2 - 0.53) \cdot V_{OL} - V_{OL}^2 \right]$$

$$4.83V_{OL}^2 - 9.17V_{OL} + 2.4 = 0$$

$$V_{OL} = 0.31V$$

$$\text{因为 } V_{OL} > \frac{(V_{GS} - V_T) E_{CN} L_N}{V_{GS} - V_T + E_{CN} L_N} = 0.25V$$

所以假设不成立, 晶体管处于饱和区, 重新解方程

$$V_{OL} = 0.3752V \quad (5+1)$$

$$V_{50\%} = \frac{V_{OH} + V_{OL}}{2} = \frac{1.2 + 0.3752}{2} = 0.7876V \quad (2)$$

$$\tau_{PHL} = \frac{C_L (V_{OH} - V_{50\%})}{I_C(\text{average})} = \frac{C_L (V_{OH} - V_{50\%})}{\frac{1}{2} (I_C|_{V_{OH}} + I_C|_{V_{50\%}})}$$

当 $V_{out} = V_{OH}$, 有 $V_{in} = V_{OH}$, 负载电阻上没有电流流过, 晶体管处于饱和区。(6+2)

$$I_C|_{V_{OH}} = I_{D(sat)} = \frac{k_n}{2} \cdot \frac{(V_{in} - V_{T,n})^2}{(V_{in} - V_{T,n}) + E_{C,n}L_n}$$

$$= \frac{98.3 \times 10^{-6} \times 0.4 \times 10}{2} \cdot \frac{(1.2 - 0.53)^2}{(1.2 - 0.53) + 0.4} = 82.5 \mu A = 0.08 mA$$

当 $V_{out}=V_{50\%}$ 时, NMOS 依然处于饱和区, 扣除负载电阻上流过的电流, $-\frac{V_{DD}-V_{50\%}}{R_L}$

$$I_C(V_{50\%}) = 41.24 \mu A \quad (5+1)$$

$$\tau_{PHL} = \frac{10 \times 10^{-15} \times (1.2 - 0.79)}{\frac{1}{2}(0.0825 + 0.04124) \times 10^{-3}} = 66.67 ps \quad (2)$$

(长沟道方法)

当 $V_{out} = V_{OL}$ 时, 假设晶体管工作在线性区

$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{k_n}{2} \frac{1}{1} [2 \cdot (V_{in} - V_{T,n}) \cdot V_{OL} - V_{OL}^2]$$

$$\frac{1.2 - V_{OL}}{10 \times 10^3} = \frac{983 \times 10^{-6}}{2} \frac{1}{1} [2(1.2 - 0.53) \cdot V_{OL} - V_{OL}^2]$$

$$V_{OL} = 0.1789 V \quad (5+1)$$

因为 $V_{OL} < (V_{GS} - V_T) = 0.67$

所以假设成立, 晶体管处于线性区

$$V_{50\%} = \frac{V_{OH} + V_{OL}}{2} = \frac{1.2 + 0.1789}{2} = 0.6895 V \quad (2)$$

$$\tau_{PHL} = \frac{C_L(V_{OH} - V_{50\%})}{I_C(average)} = \frac{C_L(V_{OH} - V_{50\%})}{\frac{1}{2}(I_C|_{V_{OH}} + I_C|_{V_{50\%}})}$$

当 $V_{out}=V_{OH}$, 有 $V_{IN}=V_{OH}$, 负载电阻上没有电流流过, 晶体管处于饱和区。 (6+2)

$$I_C|_{V_{OH}} = I_{D(sat)} = \frac{k_n}{2} \cdot \frac{(V_{in} - V_{T,n})^2}{1}$$

$$= 220.6 \mu A$$

当 $V_{out}=V_{50\%}$ 时, NMOS 依然处于饱和区

$$I_C(V_{50\%}) = 169.6 \mu A \quad (5+1)$$

$$\tau_{PHL} = 26.17ps(2)$$

b) (6)

NMOS 管工作在截止区，负载电阻对电容进行充电。

$$\frac{V_{DD} - V_{out}}{R_L} = C_L \frac{dV_{out}}{dt}$$

$$\int_{V_{OL}}^{V_{50\%}} \frac{dV_{out}}{1.2 - V_{out}} = \frac{1}{10K\Omega \cdot 1fF} \int_0^{\tau_{PLH}} dt$$

$$[-\ln(1.2 - V_{out})] \Big|_{V_{OL}}^{V_{50\%}} = \frac{1}{10^{-10}} \tau_{PLH}$$

短沟道：

$$\tau_{PLH} = 10^{-10} \left(\ln \frac{1.2-0.375}{1.2-0.79} \right) = 69.9ps(5+1)$$

长沟道：

$$\tau_{PLH} = 10^{-10} \left(\ln \frac{1.2-0.1789}{1.2-0.6895} \right) = 69.31ps(5+1)$$

6.8 (30)

短沟道情况

a) (13)

$$V_{th} = \frac{V_{T0,n} + \sqrt{\kappa} \cdot (V_{DD} - |V_{T0,p}|)}{1 + \sqrt{\kappa}} = 0.59$$

$$\frac{0.5 + \sqrt{\kappa} \cdot (1.2 - 0.48)}{1 + \sqrt{\kappa}} = 0.59$$

$$0.5 + \sqrt{\kappa} \cdot (1.2 - 0.48) = 0.59(1 + \sqrt{\kappa})$$

$$\sqrt{\kappa} = 0.692$$

$$\therefore \kappa = 0.479$$

$$\kappa = \frac{W_p}{w_n} \cdot \frac{E_{c,n} \cdot L_n}{E_{c,p} \cdot L_p} = \frac{W_p}{w_n} \cdot \frac{0.4}{1.8} = 0.479$$

$$\therefore \frac{W_p}{W_n} = 2.1568$$

积分法

$$V_{sat} = \frac{(V_{SG} + V_{t0,p})E_{c,p}L_p}{(V_{SG} + V_{t0,p}) + E_{c,p}L_p} = 0.514V$$

$$t_{rise} = t_{sat} + t_{linear} = \int_{0.12}^{0.514} \frac{C_{load}}{I_{sat}} + \int_{0.514}^{1.08} \frac{C_{load}}{I_{linear}}$$

$$I_{sat} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SG} + V_{t0,p}}{E_{cp}L_p}} \times (V_{SG} + V_{t0,p})^2$$

$$I_{linear} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SD}}{E_{cp}L_p}} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2]$$

$$W_p = 10.37\mu m^{(7+1)}$$

// 需要重新计算

$$W_n = \frac{W_p}{2.16} = 4.802\mu m^{(4+1)}$$

//需要重新计算

平均电流法

$$\tau_{rise} = \frac{C_{out}\Delta V}{I_{avg}} = 100ps$$

其中 $C_{out} = 200fF$, $\Delta V = 0.9V_{DD} - 0.1V_{DD} = 0.96V$

$$I_{avg} = \frac{1}{2} [I(V_{in} = 0, V_{out} = 0.12V) + I(V_{in} = 0, V_{out} = 1.08V)]$$

当 $V_{out}=0.12V$ 时, pMOS 处于饱和区,

$$\begin{aligned} I_{sat} &= \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SG} + V_{t0,p}}{E_{cp}L_p}} \times (V_{SG} + V_{t0,p})^2 \\ &= 8.5166 \times 10^{-6} \frac{W_p}{L_p} \end{aligned}$$

当 $V_{out}=1.08V$ 时, pMOS 处于线性区,

$$I_{linear} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SD}}{E_{cp} L_p}} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2]$$

$$= 3.4155 \times 10^{-6} \frac{W_p}{L_p}$$

代入解得

$$W_p = 12.8727 \mu m$$

$$W_n = \frac{W_p}{2.1568} = 5.9634 \mu m$$

b) (13)

$$\tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2}$$

由课本公式(6.18b)及(6.19b)，代入数据，可得：

$$\tau_{PHL} = \frac{C_{load}}{k_n} \frac{2}{E_{C,n} L_n} \frac{V_{50\%} [(V_{DD} - V_{T,n}) + E_{C,n} L_n]}{(V_{DD} - V_{T,n})^2}$$

$$\tau_{PLH} = \frac{C_{load}}{k_p} \frac{2}{E_{C,p} L_p} \frac{V_{50\%} (V_{DD} - |V_{T,p}| + E_{C,p} L_p)}{(V_{DD} - |V_{T,p}|)^2}$$

按照积分法的宽长比

$$\tau_{PLH} = 54.34 ps (5+1)$$

$$\tau_{PHL} = 114.5 ps (5+1)$$

$$\therefore \tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2} = 84.42 ps \quad 1 \text{ 分}$$

按照平均电流法的宽长比

$$\tau_{PLH} = 43.78 ps (5+1)$$

$$\tau_{PHL} = 92.19 ps (5+1)$$

$$\therefore \tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2} = 67.985 ps \quad 1 \text{ 分}$$

c) (4 分)

短沟道情况下， V_{DD} 减小时，延迟时间受到的影响不大。 V_{th} 会减小 (4)

长沟道情况

a) (13 分)

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{k_R}} \cdot (V_{DD} - |V_{T0,p}|)}{1 + \sqrt{\frac{1}{k_R}}} = 0.59$$

$$\frac{1}{k_R} = 0.479$$

$$\frac{1}{k_R} = \frac{k_p}{k_n} = \frac{\mu_p C_{ox} W_p}{\mu_n C_{ox} W_n} = \frac{46 W_p}{98 W_n} = 0.479$$

$$\therefore \frac{W_p}{W_n} = 1.02$$

积分法

$$V_{sat} = V_{SG} + V_{t0,p} = 0.72V$$

$$t_{rise} = t_{sat} + t_{linear} = \int_{0.12}^{0.48} \frac{C_{load}}{I_{sat}} + \int_{0.48}^{1.08} \frac{C_{load}}{I_{linear}}$$

$$I_{sat} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times (V_{SG} + V_{t0,p})^2$$

$$I_{linear} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2]$$

$$W_p = 8.207 \mu m (6+1)$$

$$W_n = \frac{W_p}{1.02} = 8.047 \mu m (5+1)$$

平均电流法

$$\tau_{rise} = \frac{C_{out} \Delta V}{I_{avg}} = 100 ps$$

$$\text{其中 } C_{out} = 200 fF, \quad \Delta V = 0.9V_{DD} - 0.1V_{DD} = 0.96V$$

$$I_{avg} = \frac{1}{2} [I(V_{in} = 0, V_{out} = 0.12V) + I(V_{in} = 0, V_{out} = 1.08V)]$$

当 $V_{out} = 0.12V$ 时, pMOS 处于饱和区,

$$I_{sat} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times (V_{SG} + V_{t0,p})^2$$

$$= 11.9232 \times 10^{-6} \frac{W_p}{L_p}$$

当 $V_{out}=1.08V$ 时, pMOS 处于线性区,

$$I_{linear} = \frac{1}{2} \mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SD}}{E_{cp} L_p}} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2]$$

$$= 3.6432 \times 10^{-6} \frac{W_p}{L_p}$$

代入解得

$$W_p = 9.8674 \mu m$$

$$W_n = \frac{W_p}{1.02} = 9.674 \mu m$$

b) (13 分)

$$\tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2}$$

由课本公式(6.18b)及(6.19b), 代入数据, 可得:

$$\tau_{PHL} = \frac{C_{load}}{k_n (V_{DD} - V_{T0,n})} \left[\frac{2V_{T0,n}}{V_{DD} - V_{T0,n}} + \ln \left(\frac{4(V_{DD} - V_{T0,n})}{V_{DD}} - 1 \right) \right]$$

$$\tau_{PLH} = \frac{C_{load}}{k_p (V_{DD} - |V_{T0,p}|)} \left[\frac{2|V_{T0,p}|}{V_{DD} - |V_{T0,p}|} + \ln \left(\frac{4(V_{DD} - |V_{T0,p}|)}{V_{DD}} - 1 \right) \right]$$

按照积分法的宽长比

$$\tau_{PLH} = 49.14 ps (5+1)$$

$$\tau_{PHL} = 24.87 ps (5+1)$$

$$\therefore \tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2} = 37.01 ps \quad 1 \text{ 分}$$

按照平均电流法的宽长比

$$\tau_{PLH} = 40.871 ps (5+1)$$

$$\tau_{PHL} = 20.687 ps (5+1)$$

$$\therefore \tau_p = \frac{\tau_{PLH} + \tau_{PHL}}{2} = 30.779 ps \quad 1 \text{ 分}$$

c) (4 分)

长沟道情况下:

V_{DD} 减小, I_{DS} 减小 (呈平方率关系)

V_{DD} 减小, 电压上限降低 (呈线性关系)

以上, V_{DD} 减小时, 延迟时间会增大。 V_{th} 会减小 (4)

6.9(30)

a) (15 分)

短沟道

$$V_{th} = V_{th} = \frac{V_{T0,n} + \sqrt{\kappa} \cdot (V_{DD} - |V_{T0,p}|)}{1 + \sqrt{\kappa}} = 0.58V$$

$$\kappa = \frac{W_p}{W_n} \cdot \frac{E_{c,n} \cdot L_n}{E_{c,p} \cdot L_p} = \frac{W_p}{W_n} \cdot \frac{0.4}{1.8} = 0.3265$$

$$\therefore \frac{W_p}{W_n} = 1.47(6+1)$$

$$\tau_{PHL} = \frac{C_{load}}{k_n} \cdot \frac{2}{E_{c,n} L_n} \cdot \frac{V_{50\%} [(V_{DD} - V_{T,n}) + E_{c,n} L_n]}{(V_{DD} - V_{T,n})^2}$$

$$\tau_{PHL} \leq 35ps$$

$$W_n \geq 0.59\mu m(3+1)$$

$$W_p \geq 0.867\mu m(3+1)$$

长沟道

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{k_R}} (V_{DD} - |V_{T0,p}|)}{1 + \sqrt{\frac{1}{k_R}}} = 0.58$$

$$\frac{1}{k_R} = 0.3265$$

$$\frac{1}{k_R} = \frac{k_p}{k_n} = \frac{\mu_p C_{ox} W_p}{\mu_n C_{ox} W_n} = \frac{46W_p}{98W_n} = 0.3265$$

$$\therefore \frac{W_p}{W_n} = 0.696(6+1)$$

$$\tau_{PHL} = \frac{C_{load}}{k_n (V_{DD} - V_{T0,n})} \left[\frac{2V_{T0,n}}{V_{DD} - V_{T0,n}} + \ln \left(\frac{4(V_{DD} - V_{T0,n})}{V_{DD}} - 1 \right) \right]$$

$$\tau_{PHL} \leq 35ps$$

$$W_n \geq 0.1638\mu m(3+1)$$

$$W_p \geq 0.114\mu m(3+1)$$

b) (15 分)

短沟道情况

求上升时间，输入为 0

$$V_{sat,p} = \frac{(V_{SG} + V_{t0,p})E_{c,p}L_p}{(V_{SG} + V_{t0,p}) + E_{c,p}L_p} = 0.514V$$

当 $v_{out}=0.12V$ 时，PMOS 饱和，NMOS 截止；当 $v_{out}=1.08V$ 时，PMOS 线性，NMOS 截止。

$$I_{vout=0.12} = \frac{1}{2}\mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SG} + V_{t0,p}}{E_{cp}L_p}} \times (V_{SG} + V_{t0,p})^2 = 127.75\mu A$$

$$I_{vout=1.08} = \frac{1}{2}\mu_p C_{ox} \frac{W_p}{L_p} \times \frac{1}{1 + \frac{V_{SD}}{E_{cp}L_p}} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2] = 51.232\mu A$$

$$I_{avg} = \frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08}) = 89.49\mu A$$

$$t_r = \frac{C_{load}\Delta V}{I_{avg}} = \frac{C_{load} \times (1.08 - 0.12)}{\frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08})} = 53.64ps \text{ (7+1)}$$

求下降时间，输入为 VDD

$$V_{sat,n} = \frac{(V_{GS} - V_{t0,n})E_{c,n}L_n}{(V_{GS} - V_{t0,n}) + E_{c,n}L_n} = 0.255V$$

当 $v_{out}=1.08V$ 时，PMOS 截止，NMOS 饱和；当 $v_{out}=0.12V$ 时，PMOS 截止，NMOS 线性；

$$I_{vout=1.08} = \frac{1}{2}\mu_n C_{ox} \frac{W_n}{L_n} \times \frac{1}{1 + \frac{V_{GS} + V_{t0,n}}{E_{cn}L_n}} \times (V_{GS} - V_{t0,n})^2 = 87.31\mu A$$

$$I_{vout=0.12} = \frac{1}{2}\mu_n C_{ox} \frac{W_n}{L_n} \times \frac{1}{1 + \frac{V_{DS}}{E_{cn}L_n}} \times [2(V_{GS} - V_{t0,n})V_{DS} - V_{DS}^2] = 57.90\mu A$$

$$I_{avg} = \frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08}) = 72.6\mu A$$

$$t_f = \frac{C_{load}\Delta V}{I_{avg}} = \frac{C_{load} \times (1.08 - 0.12)}{\frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08})} = 66.11ps \text{ (6+1)}$$

长沟道情况

求上升时间，输入为 0

$$V_{sat,p} = V_{SG} + V_{t0,p} = 1.2 - 0.48 = 0.72V$$

当 $v_{out}=0.12V$ 时, PMOS 饱和, NMOS 截止; 当 $v_{out}=1.08V$ 时, PMOS 线性, NMOS 截止。

$$I_{vout=0.12} = \frac{1}{2} \mu_P C_{ox} \frac{W_p}{L_p} \times \frac{1}{1} \times (V_{SG} + V_{t0,p})^2 = 178.85 \mu A$$

$$I_{vout=1.08} = \frac{1}{2} \mu_P C_{ox} \frac{W_p}{L_p} \times \frac{1}{1} \times [2(V_{SG} + V_{t0,p})V_{SD} - V_{SD}^2] = 54.65 \mu A$$

$$I_{avg} = \frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08}) = 116.75 \mu A$$

$$t_r = \frac{C_{load} \Delta V}{I_{avg}} = \frac{C_{load} \times (1.08 - 0.12)}{\frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08})} = 41.11 ps \text{ (7+1)}$$

求下降时间, 输入为 VDD

$$V_{sat,n} = V_{GS} - V_{t0,n} = 0.7V$$

当 $v_{out}=1.08V$ 时, PMOS 截止, NMOS 饱和; 当 $v_{out}=0.12V$ 时, PMOS 截止, NMOS 线性;

$$I_{vout=1.08} = \frac{1}{2} \mu_n C_{ox} \frac{W_n}{L_n} \times \frac{1}{1 + \frac{V_{GS} + V_{t0,n}}{E_{cn} L_n}} \times (V_{GS} - V_{t0,n})^2 = 240.1 \mu A$$

$$I_{vout=0.12} = \frac{1}{2} \mu_n C_{ox} \frac{W_n}{L_n} \times \frac{1}{1 + \frac{V_{DS}}{E_{cn} L_n}} \times [2(V_{GS} - V_{t0,n})V_{DS} - V_{DS}^2] = 75.26 \mu A$$

$$I_{avg} = \frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08}) = 157.68 \mu A$$

$$t_f = \frac{C_{load} \Delta V}{I_{avg}} = \frac{C_{load} \times (1.08 - 0.12)}{\frac{1}{2} \times (I_{vout=0.12} + I_{vout=1.08})} = 30.44 ps \text{ (6+1)}$$